

FM Transmitter with Raspberry Pi



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There are hundreds of ways to make an FM transmitter, and most of these designs require very basic components. These FM transmitters are easy to assemble but take a lot of time and effort in tuning. Also, these are difficult to tune very precisely.

Presented here is an FM transmitter made using Raspberry Pi that can transmit an audio in *wav* format at any desired frequency in the FM range (88-108 MHz). The best part of this project is that the audio is transmitted through a single GPIO pin of Raspberry Pi.

Working

In Raspberry Pi board, the GPIO connector has pins with different functions. The device has seven true GPIO pins, I²C interface, SPI interface, serial T_X/R_X pins and PWM pins that can be used to control external hardware. Here we have used GPIO4 (physical pin No. 7) to transmit the audio at the desired frequency. Once the program is compiled, you can run

it indicating the .wav file and transmitting frequency (explained under software sub-heading). Now if you tune an FM radio on the same frequency, you will be able to hear the audio clearly.

Modulation is done by adjusting the frequency between 100.025 MHz and 99.975 MHz using a fractional divider, which produces the audio signal. The fractional divider does not have enough resolution to produce more than 6-bit audio, but since Raspberry Pi is very fast you can do oversampling to provide about 9.5-bit audio by using 128 subsamples per real audio sample. But if you give any additional task to Raspberry Pi at the same time, the sound quality will deteriorate.

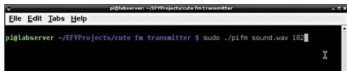
A 20cm-long wire should be connected to GPIO4 (physical pin No. 7) to transmit the audio over desired frequency in FM range.

Software

Download the source code and save it in a suitable directory as 'pifm.c'. The audio file that you want to transmit



Compile C file



Run the command

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download on www.efymag.com website.

should be placed in the same directory. The program supports only *wav* audio file format.

Compile the downloaded source code using command:

```
gcc -lm -std=c99 pifm.c -o pifm
```

This will create the executable file 'pifm'. Run this file using command:

```
$ sudo ./pifm sound.wav 100
```

Transmission frequency will be 102 MHz. You can replace '102' with the frequency of your choice. 'sound.wav' is the audio that will be transmitted over the selected frequency and kept in the same directory of source code. Tune the same frequency on your radio and you will hear sound.wav.

If you do not hear the audio after following the above-mentioned procedure, run the command 'echo "snd-bcm2835">> /etc/modules,' reboot Raspberry Pi and follow the same procedure again. The command makes the kernel module for the sound devices load on startup. ●

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